

Project: Medical Insurance Cost Prediction

Note: Uploaded cleaned R script as a second file. Structured the R script with comments so it matches the document.

Data Source for problem set: <https://www.kaggle.com/datasets/rahulvyasm/medical-insurance-cost-prediction?resource=download>

Part I: Introduction

"Medical Insurance Cost Prediction," contains information pertinent to predicting individual medical insurance costs. This data is valuable for understanding how various factors influence insurance premiums.

Background of the Business/Research Problem: In the healthcare insurance industry, accurately forecasting medical insurance costs is crucial for both insurers and policyholders. Insurers rely on precise predictions to set fair premiums, manage risk, and ensure profitability. Policyholders benefit from understanding the factors that affect their insurance costs, enabling them to make informed decisions about their healthcare plans.

Regression Problem:

- **Business/Research Question:** How can we predict the medical insurance cost for an individual based on their demographic and health-related attributes?
- **Dependent Variable (DV):** Insurance cost (charges)
- **Independent Variables (IVs):** Age, sex, BMI, number of children, smoking status, region, and any other relevant factors present in the dataset.
- **Importance in Business / Research Practice:**
 - Provide insights into how demographic and health-related factors drive healthcare expenditures.
 - Improve financial planning and risk management by forecasting claim expenses more effectively.
 - Offer competitive and fair pricing models, fostering trust and long-term customer relationships.
 - Enable healthcare researchers to identify at-risk populations and target interventions to manage insurance costs effectively.

Classification Problem:

- **Business/Research Question:** Can we classify individuals into risk categories for high vs. low insurance costs based on their demographic and health factors?
- **Dependent Variable (DV):** Risk category (binary: high-cost vs. low-cost, where a threshold on insurance cost defines the categories).
The cut-off for high-risk individuals is set at the top 25% of medical charges, categorizing

individuals with costs in this range as high-cost. This threshold ensures a clear segmentation of risk levels. Further details on the threshold selection and its justification can be found in the *Data Preparation* section.

- **Independent Variables (IVs):** Age, BMI, smoking status, number of children, region, sex, and interactions such as Smoking × BMI.
- **Importance in Business / Research Practice:** Helps insurers identify high-cost individuals for strategic planning and risk management.

General hypothesis:

Medical insurance costs are significantly influenced by an individual's demographic factors (age, sex, region) and health-related attributes (BMI, smoking status, number of children). The relationships between these variables and insurance costs may be linear or involve interaction effects, such as a combined impact of smoking and BMI or age and the number of children. By identifying these patterns, we can better understand the key drivers of insurance premiums and their implications for business and public health.

Part II: Data Preparation and EDA

Data Source and Summary

- **Dataset Source:** As mentioned earlier, the dataset used is the "Medical Insurance Cost Prediction" dataset, accessible at [Kaggle](#).
- **Observations:** The dataset contains 2,772 records. Each observation represents a unique individual, characterized by their demographic information (age, sex, region, and number of children), health-related attributes (Body Mass Index (BMI) and smoker status), and the corresponding medical insurance charges incurred by that individual.
- **Variables:**
 1. age: Age of the individual (quantitative, metric, discrete).
 2. sex: Gender of the individual (qualitative, nominal).
 3. bmi: Body Mass Index, a health metric (quantitative, metric, continuous).
 4. children: Number of children the individual has (quantitative, metric, discrete).
 5. smoker: Smoking status of the individual (qualitative, nominal).
 6. region: Geographical region of the individual (qualitative, nominal).
 7. charges: Annual medical insurance charges (quantitative, metric, continuous).
- **Completeness of Data:**
 - The whole dataset contains no missing values in any variable. This ensures that no data imputation or cleaning is necessary before analysis.

- Each variable is fully populated, making the dataset robust for both exploratory and predictive modeling purposes.

EDA: Univariate Analysis for each variable

Variable “age”

By analyzing the variable “age” it can be seen that the observations are distributed from 18 years to 64 years, 18 being minimum value and 64 being maximum value. The dataset excludes children (under 18) and seniors (65 and older), meaning the results of the following analysis cannot be generalized to these age groups. The mean of all observations within the dataset is 39.10967, which refers to the fact that the average age of people in this dataset is 39 years. In addition to that, a standard deviation of 14.08146 indicates how much the age of individuals vary around the mean age of 39 within the dataset. A lower standard deviation would mean that the data points within the dataset are close to the mean of 39 years, a higher standard deviation would mean that the data points are more spread across the minimum and maximum values of the dataset. The standard deviation helps to assess how representative the mean age is and as the standard deviation in this case is 14 years, it means that on average the age deviates 14 years from the mean of 39 years. This means that the majority of observations lie between 25 years and 53 years while assuming a normal distribution.

Table 1 - Numerical Outputs of Variable “Age”

```
> range(data$age)
[1] 18 64
> mean(data$age)
[1] 39.10967
> sd(data$age)
[1] 14.08146
```

The distribution of age is relatively balanced across most age groups, with a slight trend toward fewer individuals in older age groups. There are no potential outliers within the dataset. The youngest group (18–20 years) has 348 individuals, making it the largest group, although it spans only 3 years compared to the mostly 5-year span of other groups. The number of individuals steadily decreases across age groups, with the oldest group (61–64 years) containing only 186 individuals, nearly half the size of the youngest group.

Table 2 - Grouping Variable “Age”

```
> print(age_group_counts)
```

18-20	21-25	26-30	31-35	36-40	41-45	46-50	51-55	56-60	61-64
348	296	288	266	260	282	294	292	260	186

Besides the numerical outputs, graphical output can be used to further strengthen insights and to make them easier to understand.

Starting with the histogram of the variable age, it can be seen that all ages are well represented. While the younger generation (particularly 18–20 years) is slightly overrepresented and the oldest group (61–64) is slightly underrepresented, the relatively even distribution of individuals across the age range ensures that no single age group dominates the dataset, supporting meaningful analysis of age-related trends.

The boxplot visualization shows with its whiskers that 18 is the youngest individual and 64 is the oldest individual. It confirms that there are no outliers, as no data point lies outside of both whiskers. The median age is around 39 years as the line indicates. It further shows that the distribution is relatively symmetrical, as the median age is centrally located in the blue box. The 1st quartile (Q1) is 26 years, and the 3rd quartile (Q3) is 51 years, meaning the middle half of the dataset contains individuals aged 26 to 51 years. This results in an interquartile range (IQR) of 25 years, referring to moderate spread in the center of the dataset. These metrics further illustrate the dataset's balance, enabling robust analysis without the need for significant data cleaning.

Figure 1 - Histogram of Variable “Age”

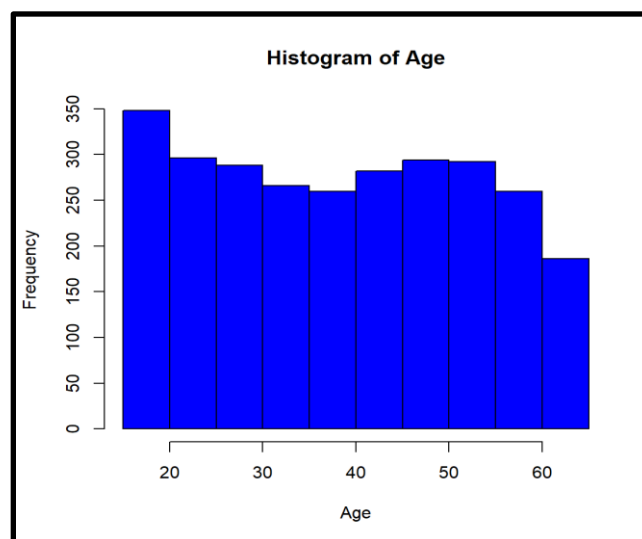
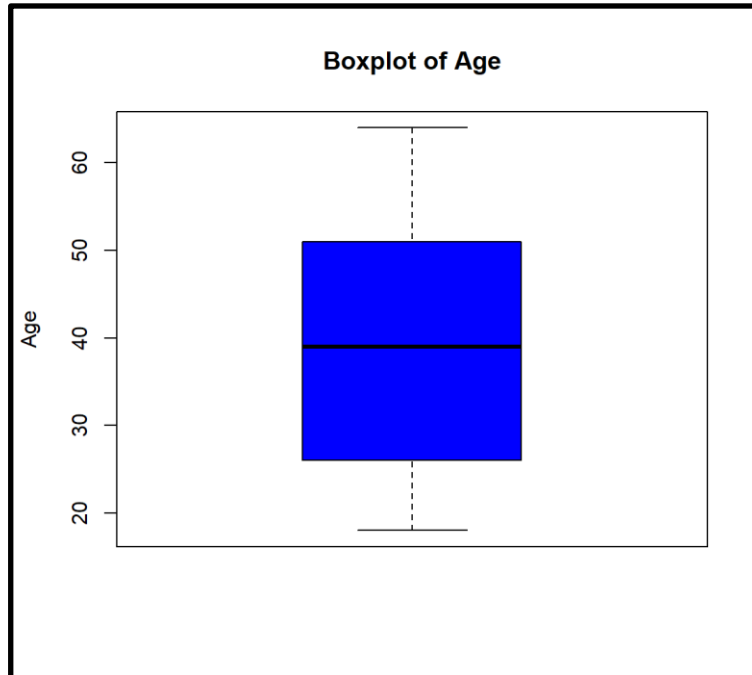


Figure 2 - Boxplot of Variable “Age”



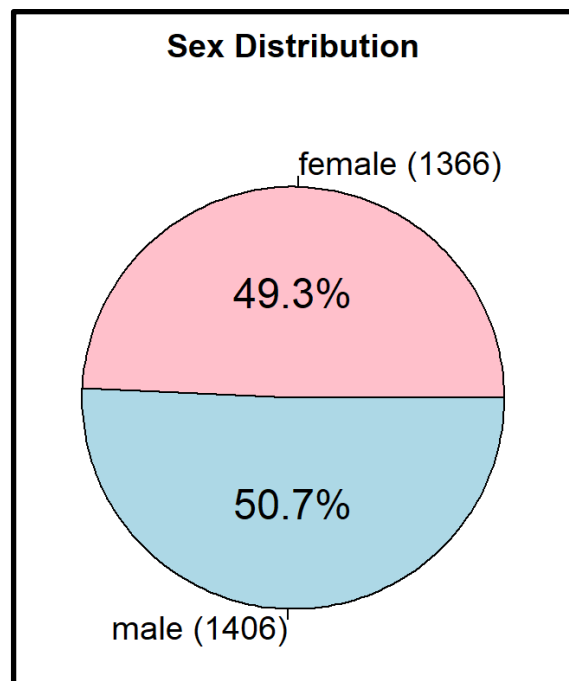
Variable “sex”

As the numerical and graphical output shows, the observations are nearly evenly split between males and females with 1406 individuals out of 2772 being males and with 1366 individuals out of 2772 being females. In percentages, this results in 50.7% of the individuals in the dataset being males and 49.3% being females. This is very good for representativeness and for the analysis as it ensures that there will be no bias towards one gender group as the dataset is almost evenly distributed. Having such a good balance between males and females is important to draw meaningful conclusions when it comes to analyzing other variables such as “smoker” or “charges” in context to the gender. Since “sex” is a qualitative variable, there is no mean, no standard deviation and also no potential outliers.

Table 3 - Numerical Output of Variable “sex”

```
> sex_counts  
  
female    male  
   1366    1406  
> sex_percentage  
  
female    male  
   49.3    50.7
```

Figure 3 - Pie Chart of Variable “sex”



Variable “bmi”

The observations in regards to variable “bmi” range from 15.96 as minimum value in the dataset to 53.13 as maximum value in the dataset. A value of 15.96 indicates underweight and a value of 53.13 indicates severe obesity according to WHO BMI classifications. The mean of all observations is approximately 30.70 and the standard deviation results in approximately 6.13. Taking into account the WHO recommendations for BMI, the average individual in this dataset is considered to be in “Obesity class 1”, meaning they are obese. Referring to the WHO, a normal body weight has a BMI between 18.5 and 24.9. An BMI above 40 indicates “Obesity Class 3” respectively

severe obesity. In addition, the standard deviation of approximately 6.13 indicates the spread in the dataset and shows that the majority of observations have a bmi between 24.57 and 36.83 (WHO, 2010).

Table 4 - Numerical Output of Variable "bmi"

```
> range(data$bmi)
[1] 15.96 53.13
> mean(data$bmi)
[1] 30.70135
> sd(data$bmi)
[1] 6.129449
```

By looking at the visualizations, the histogram underscores that most BMI values lie between 25 and 35, respectively more precisely as in the numerical output between 24.57 and 36.83. It can also be seen that only the minority has either underweight or extreme severe obesity (greater than 50), but extreme severe obesity being even less frequent than underweight.

The boxplot provides information about potential outliers. Indeed, there are several outliers within the dataset as data points are visible above the upper whisker. The spread of data can be assessed by looking at the interquartile range, which is 26.22 for the 1st quartile and 34.77 for the 3rd quartile, meaning 25% of BMI values are below 26.22 and 75% of all BMI values are below 34.77. This results in a interquartile range of 8.55, meaning that the middle 50% of BMI values in the dataset lie between 26.22 and 34.77. The smaller blue box indicates that the data points are more centered than for instance the data points regarding variable age. Therefore, the spread for bmi is smaller and values are closer to the mean in comparison to age.

Figure 4 - Histogram of Variable "bmi"

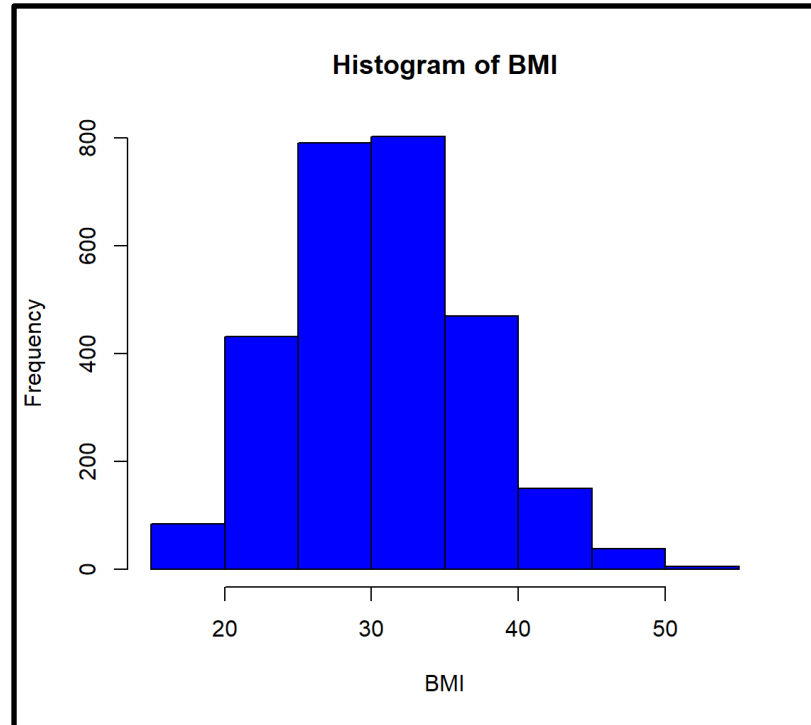
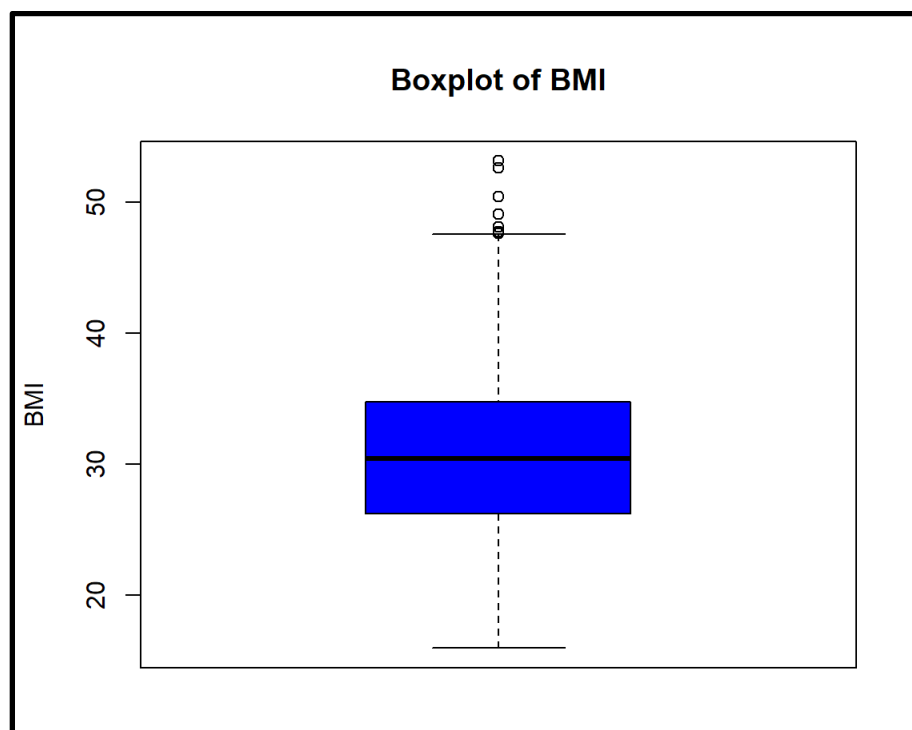


Figure 5 - Boxplot of Variable "bmi"



Variable “children”

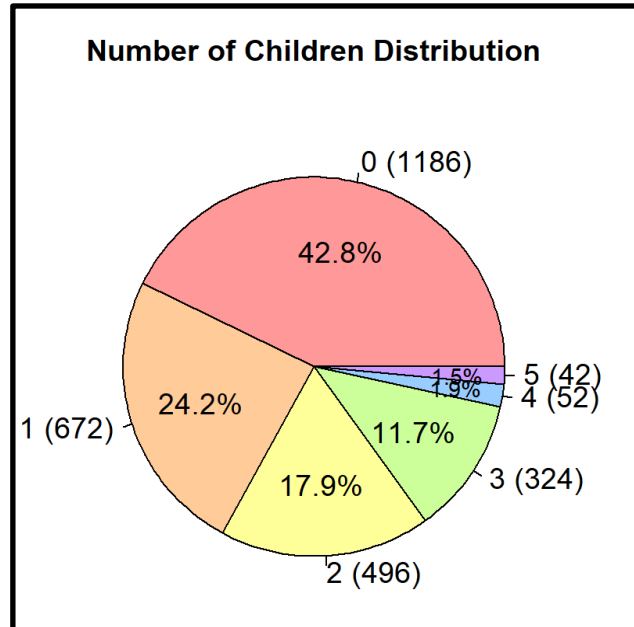
The variable “children” ranges from 0 children to having 5 children. The most individuals in this dataset have no children at all (42.8%). The frequency is steadily decreasing, meaning the more children the fewer observations resulting in only 1.9% having 4 children and only 1.5% having 5 children. Having 4 or 5 children is therefore uncommon in this dataset. It is noteworthy that after having 3 children, the frequency of observations having more than 3 children is drastically declining (from 11.7% to 1.9% and 1.5%). It can be said that larger families are less common in this dataset as only 33% have 2 or more children.

The mean of approximately 1.1 aligns with the insights above and indicates that the average individual in this dataset has only 1 child. The standard deviation is relatively moderate indicating that the majority of individuals have between 0 and 2 children, which lines up with the percentage distribution (84.90%). Regarding future analysis regarding large families of 4 or more children potentially require separate consideration as otherwise analysis could lead to not meaningful insights due to the concentration of smaller or no families.

Table 5 - Numerical Output of Variable “children”

```
> children_counts  
  
    0    1    2    3    4    5  
1186  672  496  324   52   42  
> children_percentage  
  
    0    1    2    3    4    5  
42.8 24.2 17.9 11.7  1.9  1.5  
> mean(data$children)  
[1] 1.101732  
> sd(data$children)  
[1] 1.214806
```

Figure 6 - Pie Chart of Variable “children”



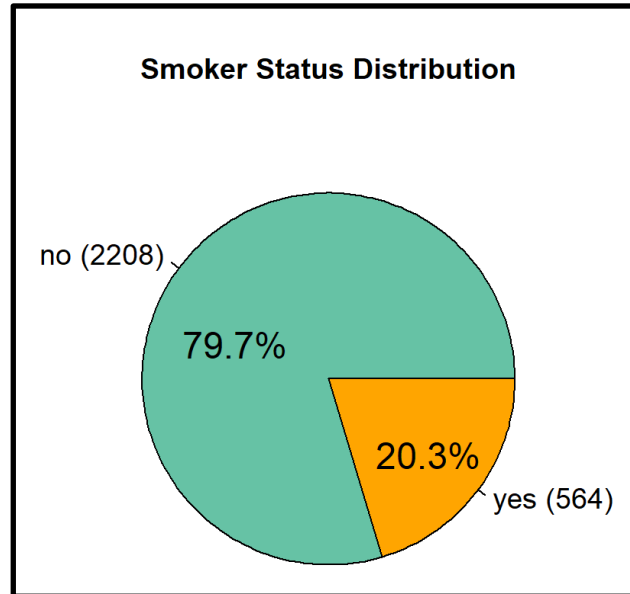
Variable “smoker”

2208 individuals of the dataset are non-smokers (79.7%) and only 564 individuals are smokers (20.3%). Therefore, distribution-wise a significant amount of the dataset is represented by non-smokers and smokers can be seen as a minority. However, as non-smokers still represent approximately 20% of the dataset, it can be used for further analysis also in combination with other variables, giving meaningful insights whether smoking influences medical costs (“charges” variable). Since “smoker” is a qualitative variable, there is no mean, no standard deviation and also no potential outliers.

Table 6 - Numerical Output of Variable “smoker”

```
> smoker_counts  
  
no    yes  
2208  564  
> smoker_percentage  
  
no    yes  
79.7  20.3
```

Figure 7 - Pie Chart of Variable “smoker”



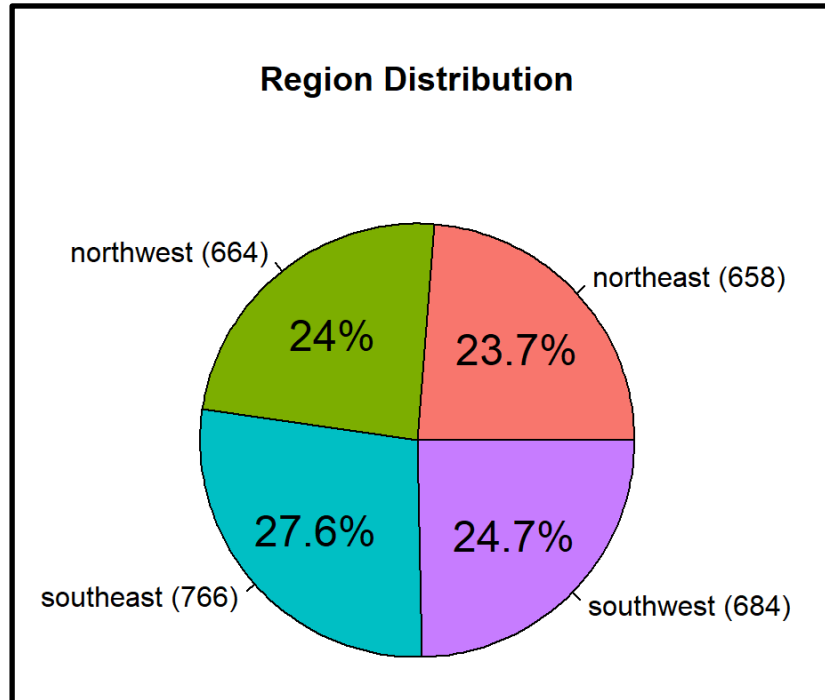
Variable “region”

658 individuals of all individuals within the dataset live in the northeast, which is equivalent to 23.7%. 664 individuals live in the northwest respectively 24%. 766 people respectively 27.6% of all individuals live in southeast and 684 individuals or 24.7% live in southwest. The largest representation within the dataset is the southeast location. On the other hand, northeast has with 658 individuals the smallest representation. Overall, the dataset is very well distributed among all 4 regions, as all 4 regions are relatively close to 25% with individual percentages ranging from 23.7% to 27.6%. This is nearly perfect for further analysis, as no region is underrepresented and future outcomes of other variables or interaction terms can be compared to different regions without distorting results. Since “region” is a qualitative variable, there is no mean, no standard deviation and also no potential outliers.

Table 7 - Numerical Output of Variable “region”

> region_counts			
northeast	northwest	southeast	southwest
658	664	766	684
> region_percentage			
northeast	northwest	southeast	southwest
23.7	24.0	27.6	24.7

Figure 8 - Pie Chart of Variable "region"



Variable "charges"

The observation regarding the variable "charges" expressed in US-Dollars ranges from 1,121.874 to 63,770.428, 1,121.874 being the minimum value and 63,770.428 being the maximum value. "charges" expresses the medical insurance cost respectively medical expenses, meaning how expensive it is to cover the medical health of the individual. The cheapest medical expenses within the dataset is therefore an individual with \$1,121.87 and the most expensive medical health expenses has an individual with \$63,770.43, indicating a very large range. The mean is equivalent to 13,261.37, meaning that on average an individual has \$13,261.37 medical expenses, which seems to be far closer to the minimum value than to the maximum value. A standard deviation of 12,151.77 provides the information that on average the medical expenses deviate \$12,151.77 from the mean expenses, resulting in the fact that the majority of medical expenses lie between \$1,109.60 and \$25,413.14. Due to the fact that \$1,109.60 is lower than the smallest value within the dataset, for this dataset it can be said that the majority of expenses lie between \$1,121.87 and \$25,413.14.

Table 8 - Numerical Output of Variable “charges”

```
> range(data$charges)
[1] 1121.874 63770.428
> mean(data$charges)
[1] 13261.37
> sd(data$charges)
[1] 12151.77
```

The histogram graphically reveals that the variable “charges” is right-skewed distributed and it underscores the insights from the numerical output, that the majority falls between \$1,121.87 and \$25,413.14. Only a very few data points represent very high medical expenses above \$50,000. By looking at the boxplot, several outliers can be identified above approximately \$35,000 meaning above the upper whisker. The further away from the majority, the less outliers are present. The 1st quartile is equal to 4,687.797 and the 3rd quartile to 16,577.78, meaning that 25% of observations are below \$4,687.80 medical expenses and 75% of observations are below \$16,577.78 medical expenses. Furthermore, this means that the interquartile range is 11,889.98 and that the middle 50% of the dataset lies between \$4,687.80 and \$16,577.78 medical expenses. The relatively small blue box of the boxplot indicates that the data points are relatively centered around the mean.

Figure 9 - Histogram of Variable “charges”

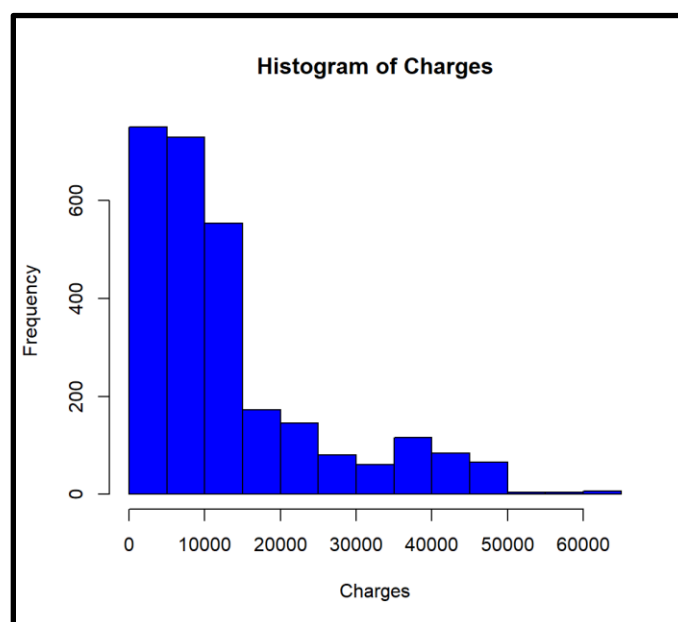
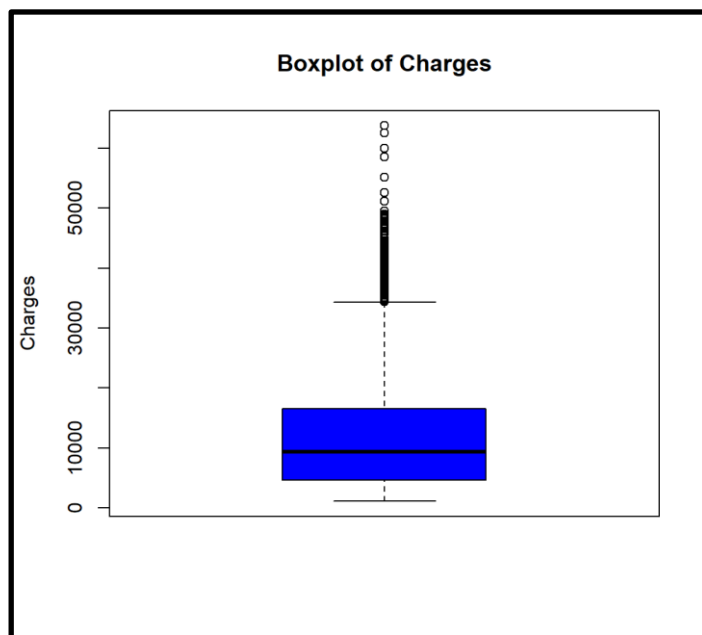


Figure 10 - Boxplot of Variable “charges”

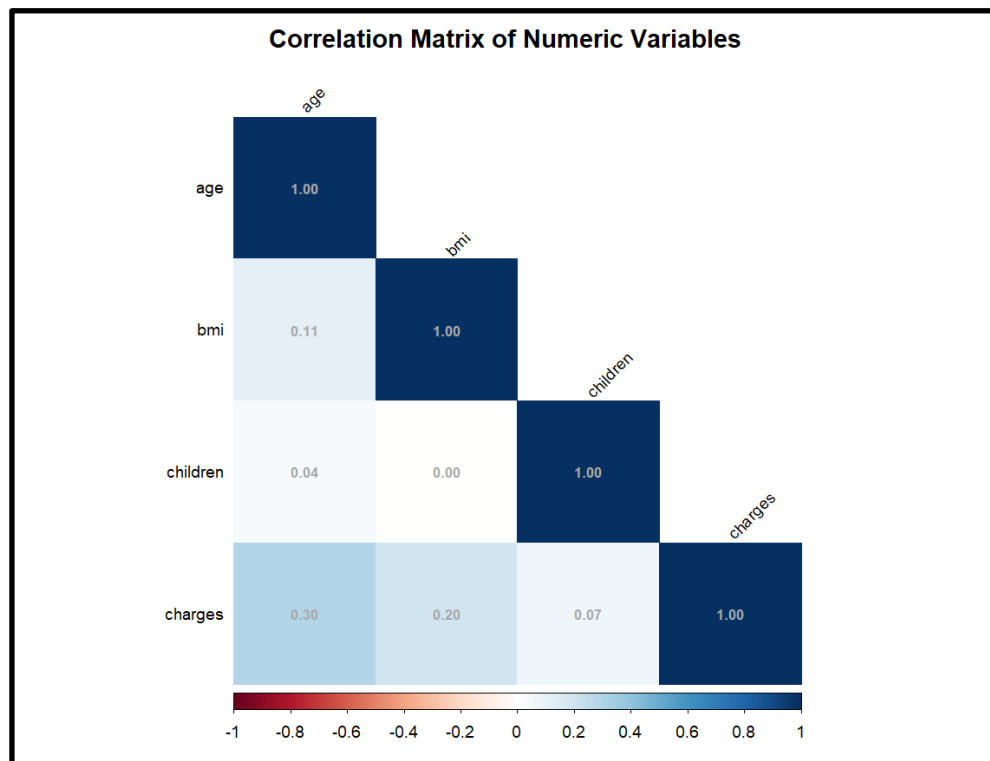


EDA: Bivariate Analysis

1. Correlation Matrix

The correlation matrix summarizes the relationships between numeric variables in the dataset, including age, bmi, children, and charges. The matrix uses Pearson's correlation coefficients, which measure the strength and direction of linear relationships between pairs of variables.

Figure 11 - Correlation Matrix of Selected Variables



The variables **“age”** and **“charges”** have a **moderate positive correlation** ($r = 0.30$), indicating that older individuals tend to incur higher medical expenses. This suggests age should be a relevant predictor in modeling insurance costs.

“bmi” and **“charges”** exhibit a **weak positive correlation** ($r = 0.20$). Although the relationship is weaker than age, it aligns with the expectation that higher BMI are also associated with increased medical expenses. A possible reason why it correlates less than age could be that also a very low BMI could increase medical expenses.

“children” and **“charges”** show a **negligible correlation** ($r = 0.07$), indicating that the number of children has little to no direct relationship with medical charges.

The weak correlations between predictors such as **“age”**, **“bmi”** and **“children”** suggest that the dataset is fairly distributed across different demographic and health-related groups. For instance, age and bmi exhibit a very weak positive correlation ($r = 0.11$), while children is nearly uncorrelated with either ($r = 0.04$ and $r = 0.00$, respectively). This independence among predictors indicates that the **dataset includes diverse individuals, ensuring that no single characteristic dominates the dataset**. This balance enhances the reliability and fairness of the subsequent analyses and modeling efforts.

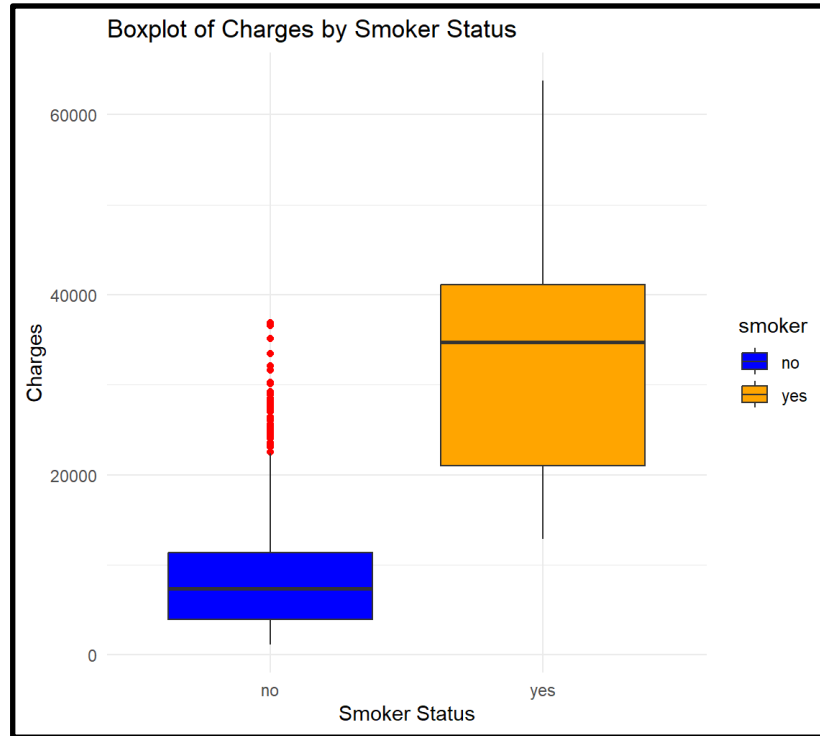
While the included predictors may be relevant for modeling, the relationships are not overly strong, **suggesting the need for further analysis**, such as regression, to explore potential nonlinearities or interactions.

2. Charges & Smoking Status and Charges, BMI & Smoking Status

The boxplot of charges and smoking status shows that the median charges for smokers are drastically higher than the median charges for non-smokers, clarifying that the variable “smoker” has a tremendous impact on the charges variable. The median charges for non-smokers are approximately below \$10,000 and for smokers around \$35,000. This shows that even without the bmi variable, the smoking status has a huge effect on a person’s medical expenses. The orange box for smokers is also significantly greater than the blue box for non-smokers, indicating that the interquartile range for smokers is larger than the interquartile range for non-smokers. This means that smokers have a higher variability around their median medical expenses, probably because if you are a smoker, you can suffer more likely from severe diseases which costs can vary depending on severity. The red-marked outliers for non-smokers represent individuals who do not smoke but have unusually high medical expenses, probably due to other severe health condition problems.

What is noteworthy is that if you have medical expenses roughly between \$20,000 and \$40,000 and you are a smoker, it is considered as usual, but if you are a non-smoker with the same amount of medical expenses, you are considered an extreme outlier. Overall, the distribution among non-smokers is compact and the distribution of smokers in regards to charges is more spread. This boxplot highlights the fact that smokers are a high risk and high cost group for medical insurers, also making it harder to predict the actual medical expenses since their distribution is more spread.

Figure 12 - Boxplot of Variables “charges” & “smoker”



The scatter plot visualizes the relationship between BMI, smoker status, and medical charges. The chart uses color coding to differentiate between smokers and non-smokers, providing insights into how these variables interact.

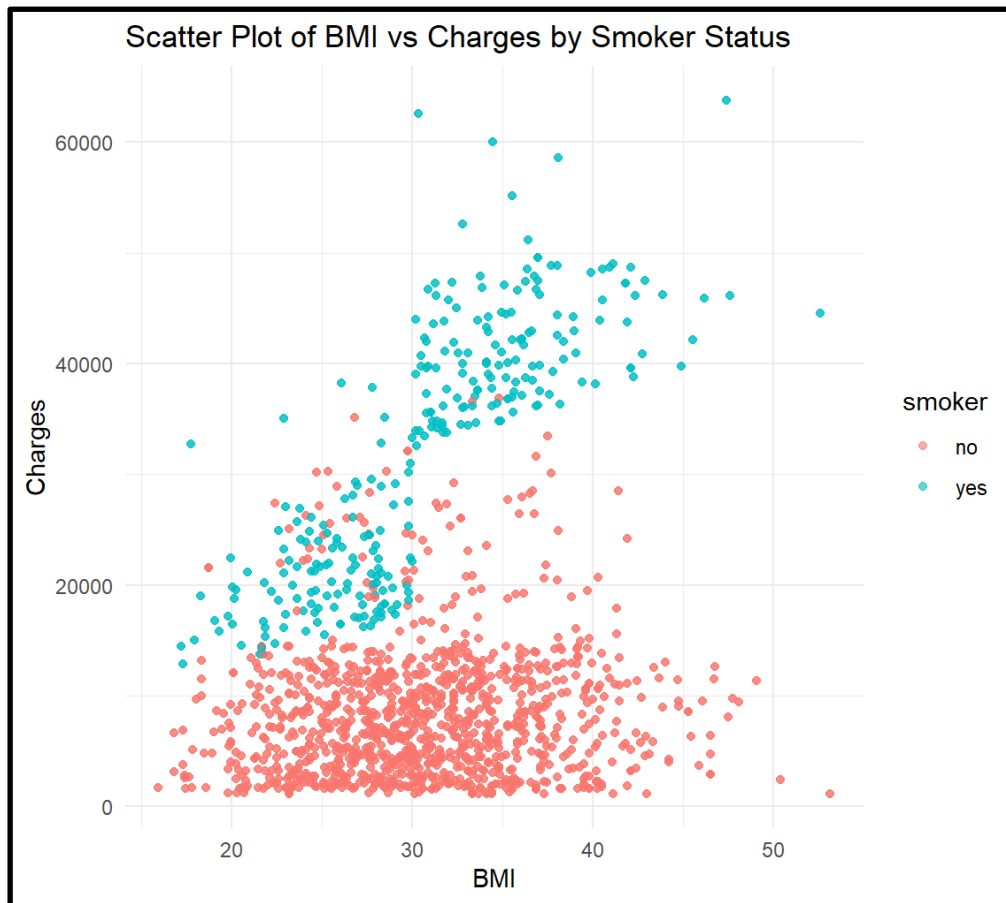
The visualization reveals that smokers in general incur higher medical expenses than non-smokers, even if the individuals have the same BMI. This strengthens the insights from the boxplot above. Very insightful is the fact that the combination of smoking and having a higher BMI above 30 leads to exorbitant higher charges while the charges for non-smoker with a BMI above 30 remains relatively stable. This means that individuals who smoke and are obese are likely to have very high medical expenses and that the bmi variable has more prediction relevance to charges variable for individuals who smoke.

Regarding outliers, non-smokers rarely exceed medical expenses of \$20,000 or more. Smokers on the other hand are more spread in general, but there are also some outliers which have medical expenses around \$60,000.

The strong interaction between smoking status and BMI in predicting charges suggests the need for including interaction terms in future regression models. This analysis

highlights the importance of considering smoking status as a key predictor and moderator in understanding medical costs.

Figure 13 - Scatter Plot of Variables “charges” & “bmi” with Categorical Variable “smoker”

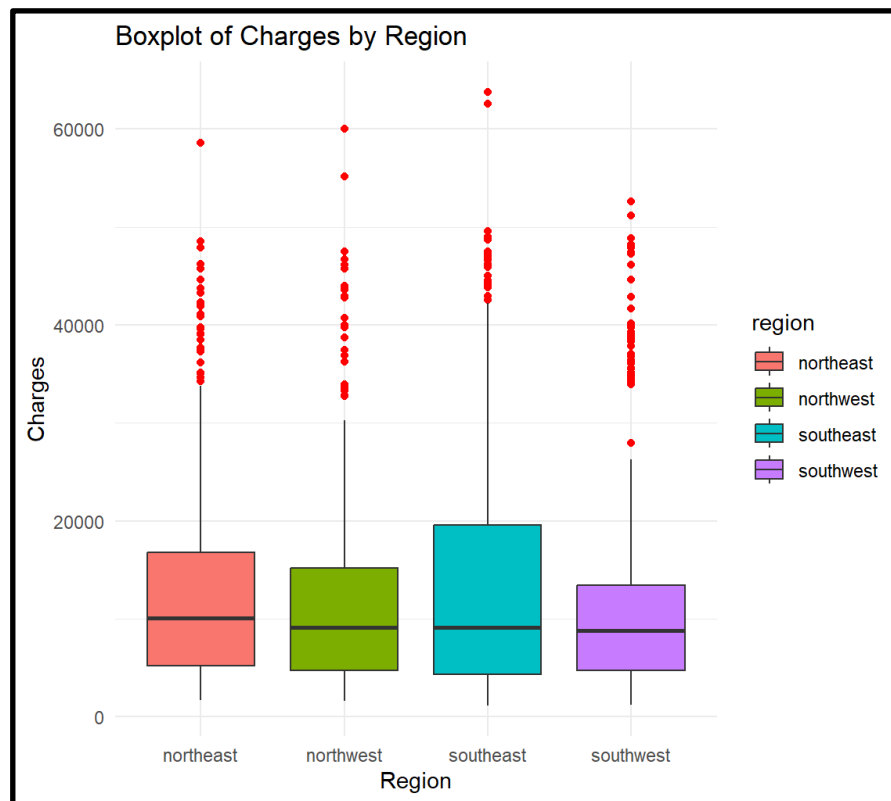


3. Region and Charges

No region seems to have a significant difference as they all have median charges around \$10,000. However, southeast has the largest interquartile range meaning that individuals living in southeast have the widest distribution in terms of medical expenses. (They vary the most). Southeast also has the largest outliers, some over \$60,000. This indicates slightly different health-related factors in southeast compared to the other regions which slightly increase medical expenses. The other three regions have relatively small distributions and are more centered around their median charges.

All regions have outliers, but again southeast having the most extreme outliers. An explanation for southeast's extreme outliers and their wider distribution might be that there are subgroups with different health condition factors within southeast, causing different medical expenses. For medical insurers it is more difficult to predict medical expenses for individuals who are located in southeast than in other regions.

Figure 14 - Boxplot of Variables “charges” & “region”



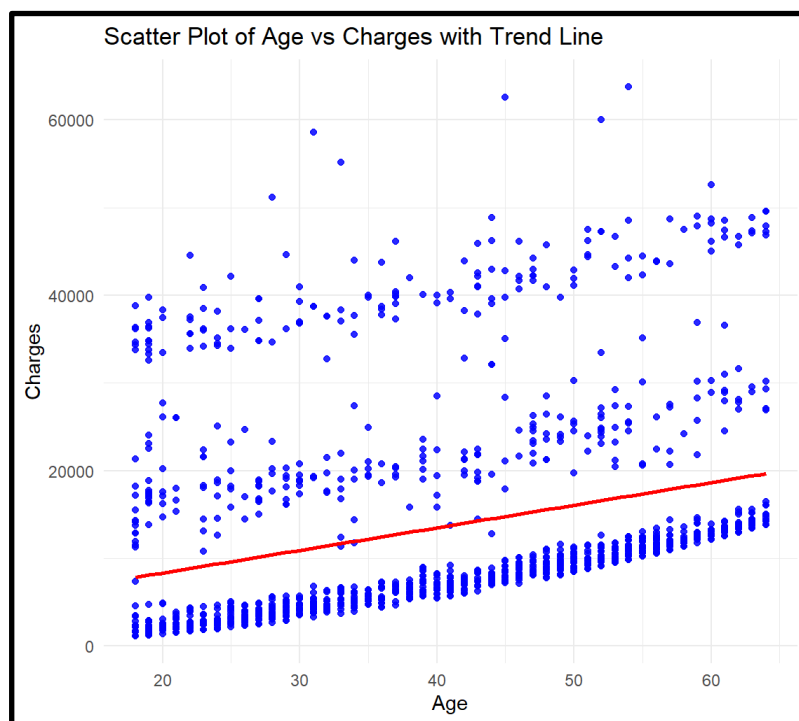
4. Age and Charges

The scatter plot emphasises that there is a positive correlation between age and charges, meaning as age increases, charges generally increase as well. This is further confirmed by the red trend line visible, although the relationship is not fully linear. Regarding distribution, individuals under 30 years usually have medical expenses below \$20,000 with low variability. In contrast, people aged older than 50 years usually have higher medical expenses and higher variability. This makes it harder to predict the exact medical expenses of older people. There are several outliers in the visualization, but the

outliers for older people are more extreme than the outliers associated with younger people.

Furthermore, although the general trend can be considered linear or close to linear, this visualization gives a hint for a non-linear pattern above the age of 50. By looking at the data points, there is a pattern of non-linear relationship that medical expenses accelerate for older individuals. This could mean that age becomes more important for prediction of medical expenses for highly-aged people. Therefore, age is a key predictor for medical cost and insurance pricing in general.

Figure 15 - Scatter Plot of Variables “charges” & “age”



Data Cleaning and Preparation

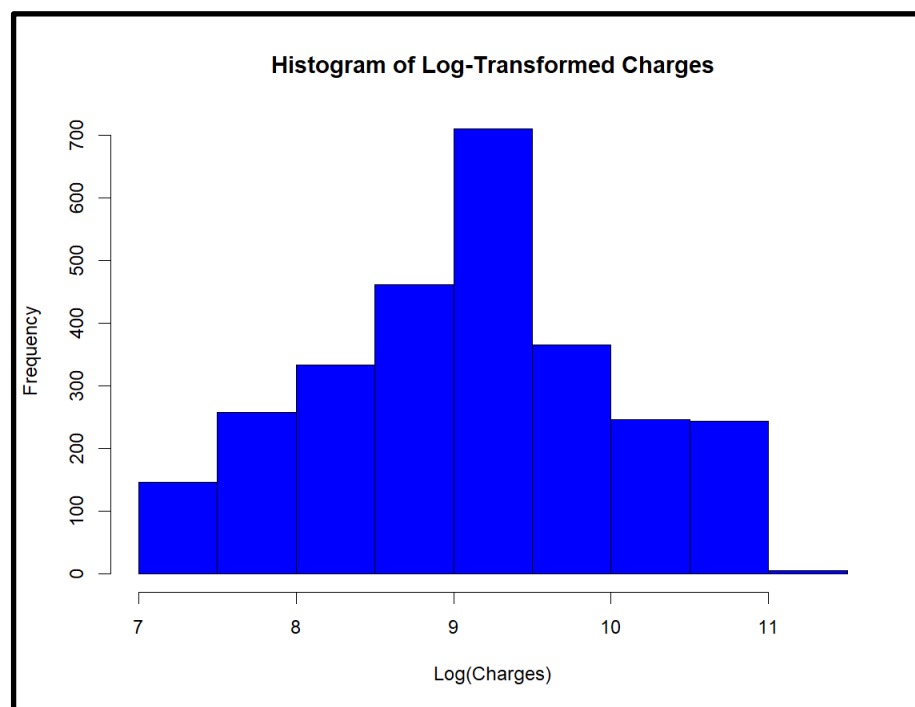
In this section, we document the comprehensive data preparation steps undertaken to optimize the dataset for regression and classification analyses. These steps address data quality issues, normalize skewed variables, engineer meaningful features, and enhance model interpretability. Each step is designed to ensure the dataset is well-suited for robust analysis and insightful modeling.

1. Considered Options

Log Transformation of Charges

The variable **charges** exhibits **significant positive skewness**, which can negatively impact linear regression assumptions, such as the normality of residuals. To address this, a natural logarithmic transformation was applied, resulting in the variable **log_charges**. The histogram of **log_charges** reveals a more symmetric distribution, which could potentially improve model stability and coefficient estimation. However, to maintain the interpretability of model predictions in terms of actual medical charges, I opted not to use **log_charges** in the final regression model. Instead, I retained this variable solely for exploratory purposes to better understand the underlying distribution of the target variable.

Figure 16 - Histogram of Log-Transformed Charges



Standardization of Numeric Variables

Predictors such as **age** and **BMI** were standardized to have a mean of 0 and a standard deviation of 1. This step ensured that all predictors were on the same scale, preventing variables with larger ranges from disproportionately influencing initial model training.

However, during the evaluation of these models, we found that the use of standardized variables made it challenging to interpret coefficients directly in the context of the original data. As a result, I decided to exclude standardized predictors from the final regression model and instead rely on the original, unscaled variables to preserve

interpretability. Standardized variables remain part of my exploratory analysis process and are documented here to demonstrate consideration of variable scaling in the modeling process.

2. Feature Engineering

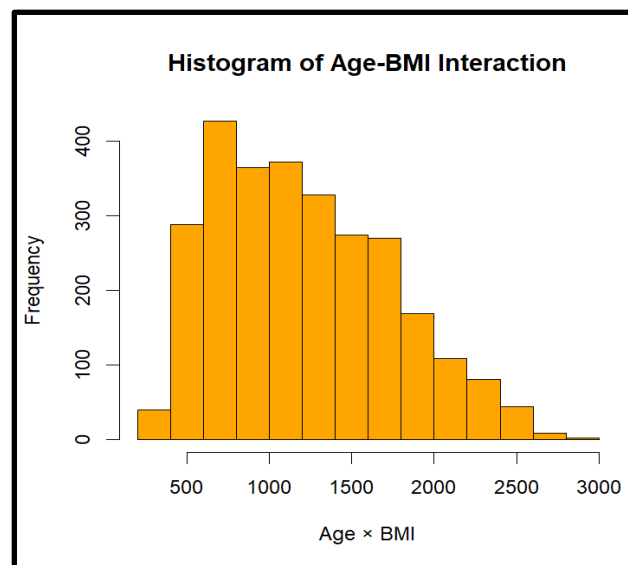
In this section, I documented all feature engineering steps undertaken, including their potential impact, and their relevance for the final model. Some features are explored but may not be included in the final model, as their utility will be determined during model selection in Part III: Analysis and Findings.

Interaction Terms

To account for potential compounding effects between predictors, I created two interaction terms:

- **Smoker × BMI (smoker_bmi_interaction):** Smokers were assigned their BMI value, while non-smokers were assigned 0. This term reflects the combined impact of smoking and BMI on charges, capturing health risks amplified by smoking among individuals with high BMI.
 - Median: 0
 - Mean: 6.28
 - Maximum: 52.58
- **Age × BMI (age_bmi_interaction):** This interaction term captures the combined influence of age and BMI on charges, which may represent age-related health risks compounded by higher BMI values.

Figure 17 - Histogram of Age-BMI Interaction

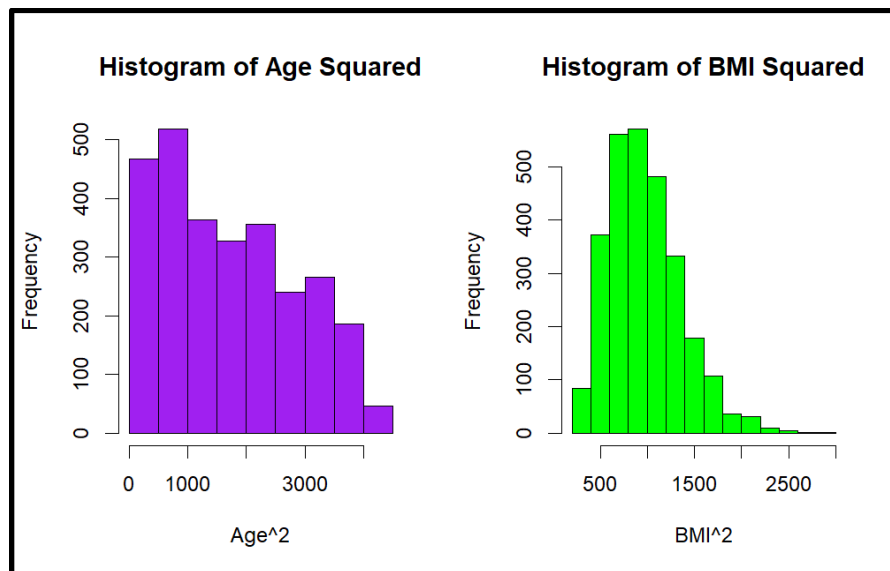


Both interaction terms were chosen based on insights from exploratory analysis and theoretical reasoning about their combined effects on medical charges. **Smoker × BMI** reflects the amplified health risks and costs associated with smoking and higher BMI as seen in the bivariate analysis about smoking. **Age × BMI** was introduced to explore whether combined age-related health risks and BMI contribute to medical charges, inspired by the strong insights gained from the **Smoker × BMI** interaction.

Polynomial Features

To model potential non-linear relationships, I introduced polynomial features for **age** and **BMI**, creating the variables **age_squared** and **bmi_squared**. These quadratic terms enhance the model's flexibility, allowing it to account for curved relationships between predictors and the dependent variable.

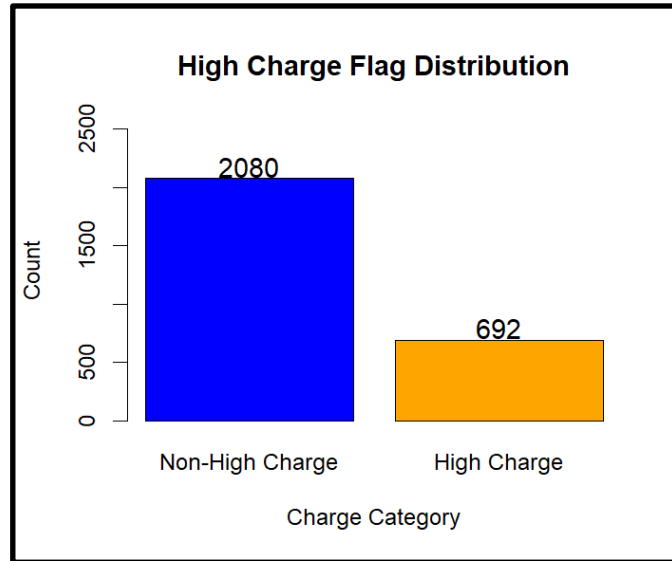
Figure 18 - Histogram of Age Squared & Histogram of BMI Squared



3. High Charge Flag

To support the classification task, I created the binary variable **high_charge** to identify individuals with medical charges in the top 25% of the dataset. The threshold for this classification was set at the 75th percentile of the **charges** variable, distinguishing the highest-cost individuals from the rest of the population.

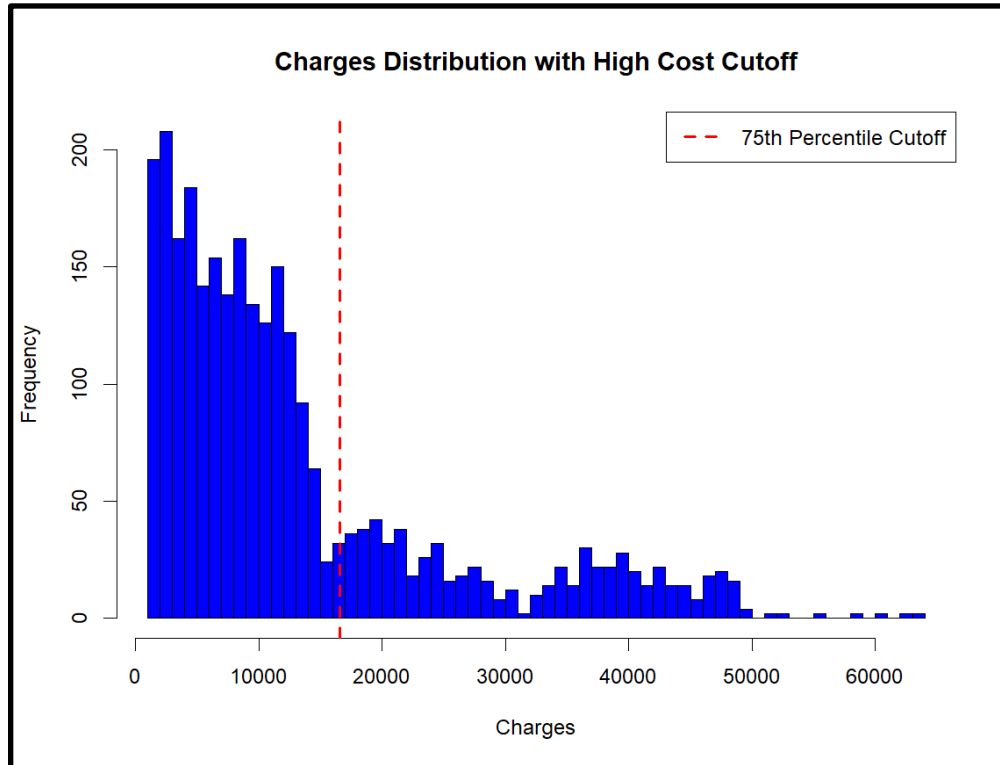
Figure 19 - High Charge Flag Distribution



The choice of the 75th percentile as the cutoff was grounded in statistical reasoning. The top quartile of charges represents a meaningful deviation from the majority of the data, as seen in the histogram of charges. This approach ensures that the high-cost group is both sufficiently distinct and large enough to enable robust model training. Additionally, the resulting segmentation aligns with standard practices for identifying and analyzing upper outliers.

- This binary segmentation provides a manageable and interpretable way to focus on the factors driving elevated medical expenses. Furthermore, it facilitates the development of classification models to predict high-cost individuals, making the dataset well-suited for this specific task.

Figure 20 - Charges Distribution with High Cost Cutoff



By isolating the top 25% of charges, I ensured the dataset captures critical insights about high-cost individuals while maintaining a balance that supports effective classification modeling. This segmentation will play a central role in subsequent analyses, particularly for understanding and predicting high medical expenses.

4. Correlation Analysis

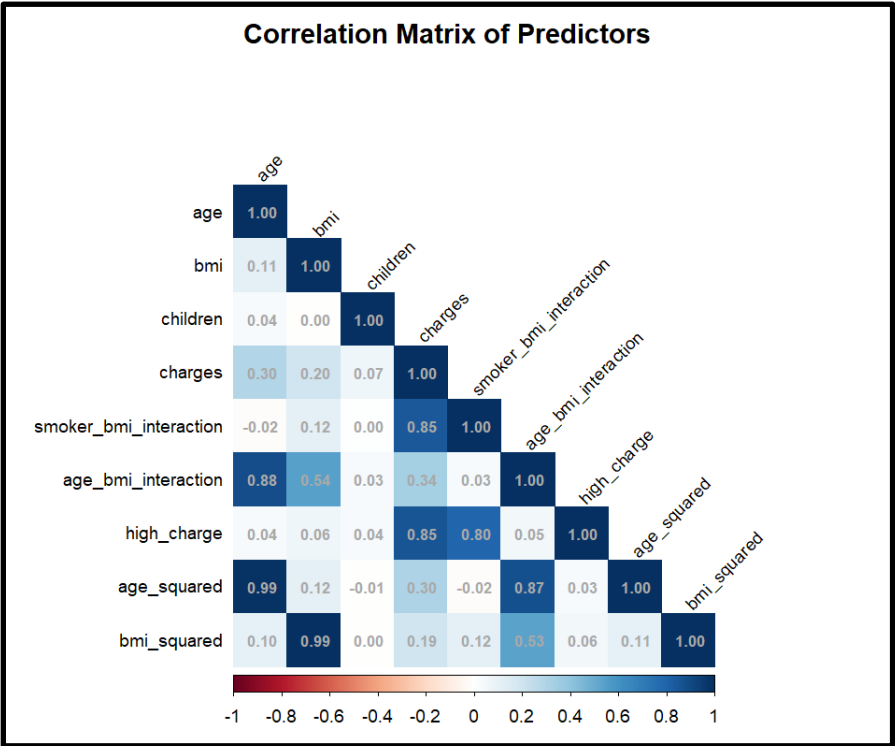
The correlation matrix was examined to identify key relationships between the predictors and the outcome variable (charges, high_charge), focusing on aspects most relevant to research questions of predicting medical costs and classifying individuals into high-cost risk categories.

interaction terms and derived features exhibit stronger patterns. The smoker $\times$ BMI interaction demonstrates a notably high correlation with charges ($r \approx 0.85$), underscoring the combined impact of smoking behavior and BMI on insurance costs. The age $\times$ BMI interaction also shows a meaningful correlation with charges ($r \approx 0.34$), suggesting that age-related health factors may compound the influence of BMI on expenses.

Finally, the binary high_charge flag, which identifies the top 25% of high-cost cases, strongly correlates with charges ($r \approx 0.85$), validating its appropriateness for classification tasks. Polynomial features (age², BMI²) show moderate correlations but

are generally overshadowed by the strength of the interaction terms. These findings guides my subsequent model-building efforts, highlighting the importance of capturing interaction effects—particularly smoker × BMI—and carefully assessing which transformations and additional features enhance predictive performance.

Figure 21 - Correlation Matrix of Predictors



This analysis ensures that the engineered features add value to the dataset and that multicollinearity is managed effectively.

Part III: Analysis and Findings

Regression problem

In this section, I focused on addressing the regression problem introduced in Part I: predicting individual medical insurance costs (charges) from a set of demographic and health-related predictors. The goal is to identify a model that provides accurate and interpretable predictions while incorporating insights gained from the Exploratory Data Analysis (Part II) and feature engineering steps.

1. Model Specification and Feature Inclusion

The predictors in regression model include original variables, transformed predictors, and interaction terms, as outlined in the table below. Detailed descriptions of these variables and their motivations can be found in the Univariate Analysis and Data Cleaning and Preparation sections.

Table 9 - Regression Model Predictors

Category	Variable	Description
Original Predictors	Age	Demographic and health-related attributes. Refer to Univariate Analysis for summaries.
	Sex	
	BMI	
	Number of Children	
	Smoking Status	
	Region	
Transformed Predictors	Age-Squared	Polynomial term for age (age^2) to capture non-linear relationships in cost prediction.
	BMI-Squared	Polynomial term for BMI (BMI^2) to model non-linear effects of BMI on costs.
Interaction Terms	Smoker-BMI-Interaction	Combines smoking status and BMI, capturing the amplified health and cost risks for smokers with higher BMI.
	Age-BMI-Interaction	Combines smoking status and BMI, capturing the amplified health and cost risks for smokers with higher BMI.

These engineered features were introduced to capture complex, non-linear relationships and potential compounding effects between predictors identified in the EDA phase (e.g., the strong combined influence of smoking status and BMI).

2. Best Subset Selection

To identify the most effective combination of predictors from original and engineered features, I applied three variable selection techniques: Best Subset, Forward Stepwise, and Backward Stepwise Selection. These methods aimed to balance predictive accuracy with model interpretability, evaluating the models based on adjusted R^2 and cross-validated RMSE.

The table below summarizes the cross-validated RMSE values for each selection method:

Model	Cross-Validation/Test RSME
Best Subset	4792.164
Forward Selection	4773.933
Backward Selection	4798.068

Despite slight differences in cross-validated RMSE, all three methods converged on the same set of predictors, indicating consistency in identifying the most relevant features for predicting medical insurance charges. While Forward Selection had the lowest RMSE, the resulting predictors align across all methods, highlighting their shared strength in feature selection.

Call:

```
lm(formula = final_subset_formula, data = transformed_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-10706.2	-1734.8	-1351.5	-614.7	30999.4

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.391e+03	2.193e+03	-1.546	0.12212
age	-8.518e+01	5.296e+01	-1.608	0.10785
sexmale	-5.030e+02	1.837e+02	-2.738	0.00622 **
bmi	5.134e+02	1.189e+02	4.319	1.62e-05 ***
children	6.786e+02	7.888e+01	8.602	< 2e-16 ***
smokeryes	-2.055e+04	1.139e+03	-18.048	< 2e-16 ***
regionnorthwest	-6.412e+02	2.649e+02	-2.420	0.01558 *
regionsoutheast	-1.240e+03	2.649e+02	-4.680	3.01e-06 ***
regionsouthwest	-1.408e+03	2.639e+02	-5.335	1.03e-07 ***
age_bmi_interaction	1.817e+00	1.073e+00	1.693	0.09058 .

```

smoker_bmi_interaction  1.448e+03  3.626e+01  39.936  < 2e-16 ***
age_squared             3.646e+00  5.628e-01   6.479  1.09e-10 ***
bmi_squared             -9.012e+00  1.821e+00  -4.949  7.91e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 4802 on 2759 degrees of freedom
Multiple R-squared:  0.8445, Adjusted R-squared:  0.8438
F-statistic: 1249 on 12 and 2759 DF, p-value: < 2.2e-16

```

To ensure the importance and relevance of all predictors included in the Forward Selection model, I further evaluated specific predictors that appeared statistically insignificant (e.g., predictors with no stars in the significance codes). Specifically, age and age_bmi_interaction were excluded to test whether their inclusion materially impacted the model's predictive performance.

A **reduced model** was created, excluding these terms, and cross-validated using the same methodology as the full model. The results showed that the reduced model had a slightly higher RMSE (4798.617) compared to the full Forward Selection model (4797.720). This minor increase in error highlighted that retaining age and age_bmi_interaction contributes to better predictive accuracy, even though these terms had marginal statistical significance individually.

In summary, the best subset method guided us toward a comprehensive, feature-rich model that outperformed simpler approaches. These findings set a strong foundation for subsequent comparisons with regularized regression methods and further refinement of my predictive modeling strategy.

In conclusion, the Forward Selection method was identified as the best approach for selecting predictors in the model, as it achieved the lowest cross-validated RMSE (4773.933), outperforming both Best Subset and Backward Selection. Despite minor differences in RMSE across the methods, all three converged on the same set of predictors, validating the robustness and relevance of the chosen variables. Further analysis of a reduced model demonstrated that retaining all predictors, including those with marginal statistical significance such as age and age_bmi_interaction, enhances the model's predictive accuracy. Thus, the Forward Selection model strikes the optimal balance between accuracy, interpretability, and comprehensive representation of demographic and health-related factors influencing medical insurance charges.

3. Comparison with Shrinkage Methods (Ridge and Lasso)

To further evaluate the robustness of this model, I tested two regularization methods: **Ridge Regression** and **Lasso Regression**. These methods penalize large coefficients

to reduce model complexity and improve generalizability. Using cross-validation, the optimal penalty parameter (λ) was determined for both methods, with 0.1 selected for each.

Ridge Regression produced a test RMSE of **5106.169**, while Lasso Regression performed better with an RMSE of **4543.447**. Despite these results, neither method outperformed the Forward Selection model, which achieved a lower RMSE of **4797.720**, indicating better predictive accuracy with the selected predictors.

The lower performance of shrinkage methods can be attributed to the strong contribution of interaction and polynomial terms included in the Forward Selection model. These predictors collectively captured nuanced relationships in the data, which could not be effectively replaced or simplified by regularization without increasing the error.

In conclusion, while Ridge and Lasso are valuable tools for simplifying models and reducing overfitting, their results in this case reaffirmed the superior performance of the Forward Selection model. This highlights the robustness of the selected predictors and validates their inclusion for accurately predicting medical insurance costs.

4. Final Model and Residual Diagnostics

The Forward Selection model was identified as the optimal regression model for predicting medical insurance costs. Among all the variable selection and shrinkage methods evaluated, Forward Selection achieved the **lowest cross-validation RMSE (4773.933)**, indicating superior predictive accuracy compared to alternative approaches. This model effectively balances complexity and performance, making it the most suitable for addressing the business objective of accurately estimating insurance charges.

Model	Cross-Validation/Test RSME
Best Subset	4792.164
Forward Selection	4773.933
Backward Selection	4798.068
Ridge Regression	5305.486
Lasso Regression	4837.977

Reduced Model	4799.470
---------------	----------

Having identified the best subset model as the final regression model, I refitted it to the entire dataset. This final model includes all the originally selected predictors, along with the specified interaction and polynomial terms. Its predictors were chosen based on the variable selection process described in the previous sections, which demonstrated that this model yields lower cross-validated errors compared to simpler models and shrinkage alternatives.

The following table presents the estimated coefficients from my final regression model. Each coefficient reflects the expected change in medical charges associated with a one-unit increase in the corresponding predictor, holding all other variables constant. The negative intercept and certain negative coefficients (e.g., **age** and **regionsoutheast**) highlight that the baseline starting point and certain regional/demographic factors can reduce costs, while positive coefficients (e.g., **children**, **smoker_bmi_interaction**) confirm variables that substantially increase the predicted charges.

Table 10 - Estimated Coefficients of Final Regression Model

Term	Estimate	Interpretation
(Intercept)	-3390.995	Baseline when all predictors are at zero/reference level
age	-85.184	Slight decrease per year of age at mean levels, adjusted by other terms
sexmale	-503.001	Males have lower charges compared to females, all else equal
bmi	513.435	Increasing BMI raises costs, with non-linear effects captured elsewhere
children	678.558	Each additional child increases expected medical charges
smokeryes	-20551.460	Large negative baseline shift, offset by interaction effects (see below)

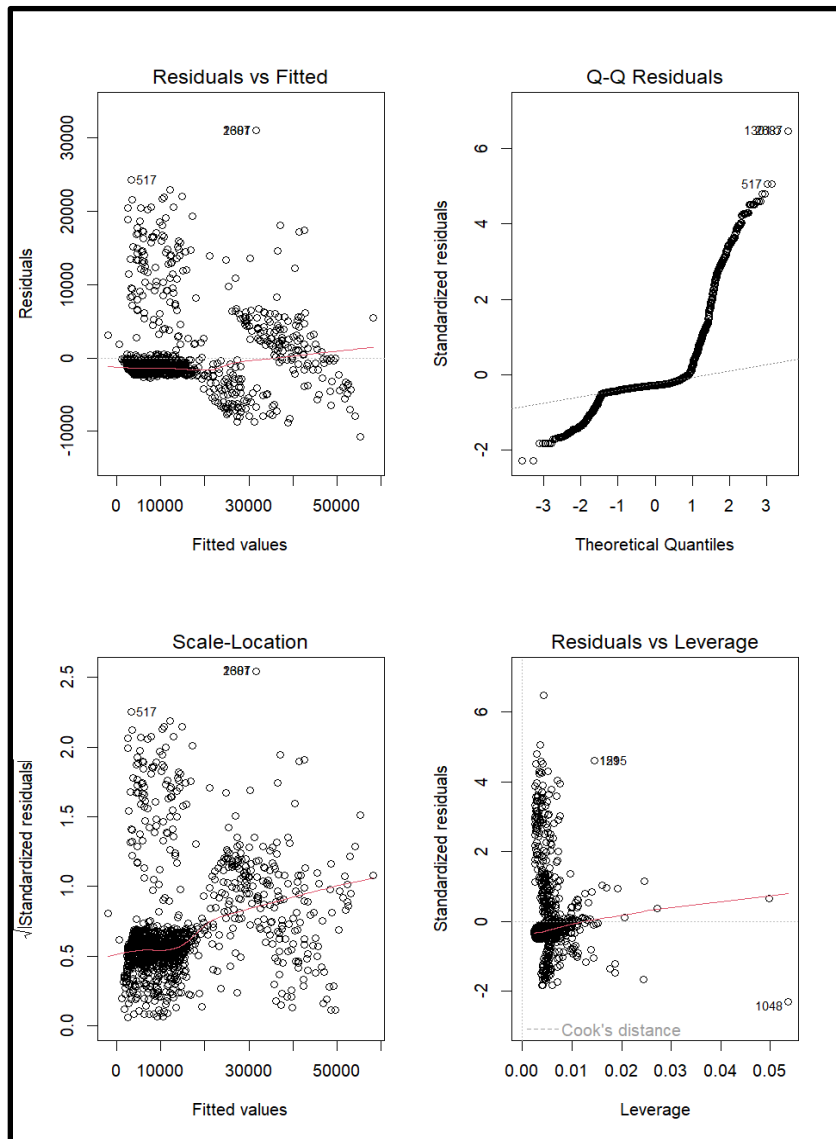
regionnorthwest	-641.161	Northwest region slightly reduces charges relative to northeast
regionsoutheast	-1239.550	Southeast region further reduces charges relative to northeast
regionsouthwest	-1407.799	Southwest region reduces charges relative to northeast even further
age_bmi_interaction	1871	Interaction: With higher BMI, effect of age becomes more positive
smoker_bmi_interaction	1448.098	Smokers with higher BMI see a large increase in costs
age_squared	3646	Older ages incur higher charges at an increasing rate (non-linear)
bmi_squared	-9.012	Non-linear term for BMI; after a point, very high BMI reduces predicted increase

To assess model adequacy and adherence to regression assumptions, I examined the residuals visually. The diagnostic plots show:

- **Residuals vs. Fitted:** Shows a pattern indicating that residuals may not be entirely random. Some curvature suggests that the linear model may not fully capture the complexity of the data.
- **Q-Q Plot:** The residuals deviate from the theoretical normal line, especially in the upper tail, indicating that the normality assumption may be violated.
- **Scale-Location Plot:** Suggests non-constant variance (heteroscedasticity), as the spread of residuals increases with fitted values.
- **Residuals vs. Leverage:** A few points have higher leverage and potentially more influence on the fitted model. These should be inspected further.

While final model offers improved predictive performance relative to simpler models or shrinkage alternatives, the residual diagnostics suggest that further improvements (e.g., transformations, alternative modeling techniques, or robust regression approaches) could be considered to better meet regression assumption

Figure 22 - Diagnostic Plots



5. Summary of Regression Findings

The analysis shows that smoking status, with a coefficient of approximately -20,550, is by far the strongest predictor of higher medical insurance costs, reflecting the substantial added expenses associated with smoking-related health risks. When considered in conjunction with the strong interaction term between smoking and BMI (roughly +1,448), it becomes clear that smokers with higher BMI values face disproportionately greater costs. This is further underscored by the fact that, on average, each additional unit of BMI adds about +513 to an individual's expected

charges. These figures suggest that interventions aimed at reducing smoking prevalence and managing weight could lead to significant cost savings.

Beyond these key behavioral factors, having more children also drives costs upward, with each additional child increasing charges by roughly +678. Regional differences are evident as well, as living in certain areas leads to lower charges: for instance, residing in the Southwest corresponds to about 1,408 less in charges than in the baseline region. Similar though somewhat smaller reductions are seen in the Southeast (-1,240) and Northwest (-641), suggesting that localized healthcare practices, pricing standards, and resource availability play a role in shaping insurance expenses.

While the direct linear effect of age (approximately -85 per year) is not statistically significant, the model does uncover meaningful non-linear patterns. Specifically, older ages come with accelerating costs, as indicated by the positive age-squared term (+3.65), while very high BMI levels produce diminishing returns, captured by a negative BMI-squared coefficient (-9.01). Taken together, these non-linear and interaction effects highlight the complexity of cost determination and reaffirm the need for tailored, data-driven strategies.

Overall, by using a robust variable selection process, I identified a model that not only predicts insurance costs with reasonable accuracy but also pinpoints the most critical factors influencing these costs. Behavioral factors like smoking and BMI clearly stand out, but demographic, family, and regional considerations all contribute to the final cost structure. Although the model's residual diagnostics suggest that further refinements might still improve its predictive power, these results provide a meaningful foundation for making informed decisions about cost management and targeted interventions

Classification problem

Original Predictors:

- Age, sex, BMI, number of children, smoking status, region.

Transformed Predictors:

- Log-transformed charges (log_charges) to normalize the distribution.
- Polynomial terms: Age-squared and BMI-squared for nonlinear relationships.
- Interaction terms: Smoker-BMI interaction (smoker_bmi_interaction), capturing combined effects.

Interaction Terms:

- Smoker $\times$ BMI and Age $\times$ BMI.

These terms are well-prepared, standardized, and validated for analysis.

1. Logistic Regression Analysis

Summary:

Call:

```
glm(formula = high_charge ~ ., family = binomial, data = train_data[,  
  c(predictors, response)])
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.16025	0.14065	-15.359	< 2e-16	***
age	0.32648	0.09167	3.561	0.000369	***
bmi	0.21257	0.09910	2.145	0.031951	*
smoker_bmi_interaction	0.20381	0.01063	19.179	< 2e-16	***
age_squared	-0.14568	0.10024	-1.453	0.146144	
bmi_squared	-0.29993	0.07830	-3.830	0.000128	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2491.2 on 2217 degrees of freedom
Residual deviance: 1085.9 on 2212 degrees of freedom
AIC: 1097.9

Number of Fisher Scoring iterations: 6

Confusion Matrix:

Reference

Prediction	0	1
0	407	32
1	8	107

Accuracy : 0.9278

95% CI : (0.903, 0.9479)

No Information Rate : 0.7491

P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.7962

McNemar's Test P-Value : 0.0002762

Sensitivity : 0.9807

Specificity : 0.7698

Pos Pred Value : 0.9271

Neg Pred Value : 0.9304

Prevalence : 0.7491

Detection Rate : 0.7347

Detection Prevalence : 0.7924

Balanced Accuracy : 0.8753

'Positive' Class : 0

Predictors:

- age: Positive coefficient (0.32648), significant ($p = 0.000369$)
- bmi: Positive coefficient (0.21257), significant ($p=0.031951$)
- smoker_bmi_interaction: Positive and highly significant ($p<2e-16$)
- age_squared: Not significant ($p=0.146144$)
- bmi_squared: Negative and significant (-0.29993 , $p=0.000128$)

Model Performance Metrics:

- **Null Deviance:** 2491.2 (baseline model with no predictors).
- **Residual Deviance:** 1085.9 (after including predictors). A significant reduction, indicating the predictors substantially improve the model.
- **AIC:** 1097.9, suggesting the model's relative quality (lower is better).

The logistic regression model provides valuable insights into the classification of individuals as high-cost or low-cost based on demographic and health-related attributes. Key predictors such as age, BMI, smoker-BMI interaction, and BMI-squared significantly impact classification outcomes. The smoker-BMI interaction emerges as the most influential factor, highlighting the compounded risks of smoking and higher BMI levels. Nonlinear effects, captured by BMI-squared, reveal diminishing probabilities of classification as high-cost at extreme BMI levels. The reduction in residual deviance from 2491.2 to 1085.9 and an AIC of 1097.9 indicate a well-fitting model that substantially explains variance compared to the null model.

	Reference: 0	Reference: 1
Predicted 0	407	32
Predicted 1	8	107

The model achieves an accuracy of 92.78%, with particularly strong sensitivity at 98.07%, meaning it effectively identifies low-cost individuals. Specificity, at 76.98%, is slightly lower, reflecting some challenges in distinguishing high-cost individuals. The balanced accuracy of 87.53% and high precision (92.71%) underscore the model's ability to maintain balanced classification performance despite the class imbalance. These results suggest that the model is reliable in identifying both high-cost and low-cost individuals with a strong emphasis on minimizing false negatives.

The findings have significant implications for insurers and policymakers. By identifying critical cost-driving factors such as smoking and BMI, insurers can develop targeted strategies to manage risk, such as incentivizing healthier behaviors through tailored wellness programs. The high sensitivity ensures accurate identification of most low-cost individuals, while the specificity, though slightly weaker, still reasonably identifies high-cost individuals. This makes the model a practical tool for risk segmentation, pricing optimization, and resource allocation.

2. LDA results and analysis

Summary

```
Call:
lda(high_charge ~ ., data = train_data[, c(predictors, response)])

Prior probabilities of groups:
      0      1
0.7506763 0.2493237
```

Group means:

	age	bmi	smoker_bmi_interaction	age_squared	bmi_squared
0	-0.03175847	-0.03905543	0.4197177	0.9976514	1.0245776
1	0.06451143	0.10906759	23.6926763	0.9895413	0.9071844

Coefficients of linear discriminants:

	LD1
age	0.09547235
bmi	-0.03166833
smoker_bmi_interaction	0.13090741
age_squared	-0.03821271
bmi_squared	-0.10861437

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	407	31
1	8	108

Accuracy : 0.9296

95% CI : (0.905, 0.9495)

No Information Rate : 0.7491

P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.8018

McNemar's Test P-Value : 0.000427

Sensitivity : 0.9807

Specificity : 0.7770

Pos Pred Value : 0.9292

Neg Pred Value : 0.9310

Prevalence : 0.7491

Detection Rate : 0.7347

Detection Prevalence : 0.7906

Balanced Accuracy : 0.8789

'Positive' Class : 0

Prior Probabilities of Groups:

- Class 0 (low-cost): 75.07%
- Class 1 (high-cost): 24.93%

This indicates an imbalance in the dataset, with a higher proportion of low-cost individuals.

Group Means:

- Variables like **smoker_bmi_interaction** show significant differences between the groups, highlighting its strong predictive power.
 - **Low-cost (Class 0):** Mean = 0.4197
 - **High-cost (Class 1):** Mean = 23.6927

Other variables such as age and BMI also show differences, though they are less pronounced, yet still important for classification.

Coefficients of Linear Discriminants:

- **smoker_bmi_interaction:** The highest positive weight (0.1309), underlining its significant contribution to differentiating between the two classes.
- **bmi_squared:** A negative weight (-0.1086), indicating a nonlinear effect of BMI on the classification.
- **age_squared:** A negative weight (-0.0382), suggesting a smaller impact compared to the smoker-BMI interaction.

The Linear Discriminant Analysis (LDA) model delivers strong classification performance for distinguishing between high-cost and low-cost individuals. Key predictors, including age, BMI, smoker-BMI interaction, and quadratic terms for age and BMI, contribute to the model's effectiveness. Among these, the smoker-BMI interaction shows the greatest impact, emphasizing the significant combined effect of these variables on medical costs. The group means for predictors, particularly the stark difference in smoker-BMI interaction between high- and low-cost groups, illustrate the model's ability to leverage these relationships for classification.

	Reference: 0	Reference: 1
Predicted 0	407	31
Predicted 1	8	108

The model achieves an overall accuracy of 92.96%, with a balanced accuracy of 87.89%, reflecting robust and reliable classification. Sensitivity is particularly strong at 98.07%, ensuring that most low-cost individuals are identified correctly. Specificity is

slightly better than logistic regression at 77.70%, demonstrating an improved ability to identify high-cost individuals. These metrics, coupled with high precision (92.92%) and negative predictive value (93.10%), highlight the model's effectiveness in handling imbalanced data while minimizing misclassification.

The LDA model provides actionable insights for risk segmentation and strategic decision-making. Its strong performance in identifying high-cost individuals supports targeted intervention strategies, such as tailoring premiums or offering wellness programs for at-risk populations. The ability to balance sensitivity and specificity makes it a practical tool for insurers aiming to optimize resource allocation and minimize financial risks. By utilizing linear relationships and group mean differences effectively, LDA enhances understanding of cost-driving factors, offering valuable guidance for policy and business strategies.

3. KNN results and analysis:

Optimal K=1

Confusion Matrix and Statistics

		Reference	
Prediction		0	1
0	410	7	
1	5	132	

Accuracy : 0.9783

95% CI : (0.9625, 0.9888)

No Information Rate : 0.7491

P-Value [Acc > NIR] : <2e-16

Kappa : 0.9421

McNemar's Test P-Value : 0.7728

Sensitivity : 0.9880

Specificity : 0.9496

Pos Pred Value : 0.9832

Neg Pred Value : 0.9635

Prevalence : 0.7491

Detection Rate : 0.7401

Detection Prevalence : 0.7527

Balanced Accuracy : 0.9688

'Positive' Class : 0

The K-Nearest Neighbors (KNN) model with K=1 delivers the best classification accuracy of 97.83%, demonstrating its ability to distinguish between high-cost and low-cost individuals accurately. With a sensitivity of 98.80% and specificity of 94.96%, KNN performs exceptionally well in identifying both low-cost and high-cost individuals, outperforming logistic regression and LDA in these metrics. The model also shows a balanced accuracy of 96.88%, and its positive predictive value (98.32%) and negative predictive value (96.35%) confirm its reliability in making correct classifications with minimal misclassifications.

	Reference: 0	Reference: 1
Predicted 0	410	7
Predicted 1	5	132

The strong performance of the KNN model has important implications for insurers and healthcare providers, as it offers a highly effective tool for risk management and resource allocation. By accurately identifying high-cost individuals, insurers can optimize pricing strategies, target wellness programs, and better allocate resources to mitigate financial risk. Its superior performance in both sensitivity and specificity makes it an invaluable model for businesses seeking to manage healthcare costs effectively while ensuring fair premium pricing and intervention strategies.

Comparison of Models

Metric	Logistic Regression	LDA	KNN (K=1)
Accuracy	92.78%	92.96%	97.83%
Sensitivity	98.07%	98.07%	98.80%
Specificity	76.98%	77.70%	94.96%

Balanced Accuracy	87.53%	87.89%	96.88%
Kappa	0.7962	0.8018	0.9421

KNN outperforms both logistic regression and LDA across all key metrics, particularly in specificity and balanced accuracy. With K=1, KNN achieves the highest classification accuracy, demonstrating its ability to capture complex, nonlinear relationships and handle intricate decision boundaries. This makes it especially effective in identifying high-cost individuals, which is crucial for insurers aiming to manage risk. Its high specificity ensures accurate identification of high-cost cases, while its sensitivity remains excellent, minimizing misclassification. The model's reliable predictions make it a powerful tool for strategic planning and targeted interventions. Given its superior performance in both sensitivity and specificity, **KNN (K = 1) is the optimal model for this classification task.**

Business and Research Findings Based on KNN (Optimal Model)

Using the optimal classification model (K=1 in KNN), business and research questions outlined in **Part I** can be addressed.

KNN Model Performance and Key Insights

The KNN model demonstrated exceptional predictive accuracy, making it the best-performing model for classifying high-cost and low-cost individuals. Key insights from the model include:

- **High Predictive Accuracy:** Achieved 97.83% accuracy and 96.88% balanced accuracy, proving the model's capability to distinguish between the two categories effectively.
- **High Sensitivity:** With a sensitivity of 98.80%, the model correctly identifies the majority of low-cost individuals, minimizing the number of false negatives.
- **High Specificity:** At 94.96%, the model correctly identifies high-cost individuals, ensuring that those most likely to incur significant claims are accurately classified.
- **Minimal Misclassification:** The high precision (98.32%) and negative predictive value (96.35%) indicate that the model provides reliable predictions, with few false positives or negatives.

Important Predictors in KNN Model

- The KNN model identified several key predictors that significantly impact medical costs. These include:
- **Smoker × BMI Interaction:** The interaction between smoking and BMI is the most influential predictor. It shows how smoking amplifies the effect of BMI on healthcare costs, highlighting the compounded risk associated with unhealthy lifestyles.
- **BMI and Age (Linear & Squared Terms):** BMI and its squared term reveal a nonlinear relationship, with extremely high BMI levels diminishing the likelihood of being classified as high-cost. Age and its quadratic term also show subtle effects, indicating that both age and BMI interact in more complex ways than a simple linear relationship.
- **Lifestyle and Demographic Factors:** The combination of smoking, BMI, and age reinforces the idea that demographic factors, along with lifestyle choices, are the key drivers of medical insurance costs.

Implications for the Healthcare Insurance Industry

The findings from the KNN model offer actionable insights for the healthcare insurance industry, specifically in pricing strategies, resource allocation, and risk management:

- **Premium Pricing:** Incorporating lifestyle factors (such as smoking and BMI) along with demographic factors (like age) into premium pricing models allows insurers to set more accurate and equitable premiums.
 - This ensures that premiums reflect an individual's true risk, contributing to fairer pricing while maintaining profitability.
 - Smoking cessation programs and weight management interventions can be incentivized through premium discounts for healthier policyholders.
- **Targeted Wellness Programs:** Insurers can offer targeted wellness programs to high-cost individuals identified through the model. For example, smokers with high BMI could be enrolled in personalized wellness initiatives aimed at reducing their healthcare costs.
- **Resource Allocation:** By identifying regions or age groups with higher predicted costs, insurers can prioritize these segments for outreach programs, balancing their portfolio and minimizing long-term financial risks.
- **Policyholder Education:** Educating policyholders about the key cost-driving factors, like smoking and BMI, empowers them to make informed decisions that improve their health and reduce insurance costs.

Conclusion

The KNN model, with its high accuracy and balanced performance, is highly effective in classifying individuals into high-cost and low-cost categories for medical insurance. It provides valuable insights into key predictors like smoking, BMI, and their interactions, aiding in more accurate pricing, resource allocation, and targeted wellness programs. The model's ability to minimize misclassification, particularly for high-cost individuals, makes it a vital tool for risk management. Additionally, the use of non-parametric methods highlights the importance of capturing complex relationships within healthcare data, opening opportunities for future research to refine predictions and improve cost management strategies for insurers and policyholders.

Part IV: Conclusions and Recommendations

The comprehensive analysis of medical insurance cost data not only identifies the primary factors influencing individual expenses but also distinguishes high-cost individuals from those likely to incur lower charges. Using regression techniques, I found that smoking status is the strongest single predictor of medical costs, with smokers incurring expenses that can average around 20,550 USD more than comparable non-smokers. This effect is compounded by body mass index (BMI), as the combination of smoking and high BMI values leads to disproportionately high costs, underscoring the interplay between lifestyle choices and financial risk. Beyond these factors, each additional child increases expenses by about 678 USD, and certain regions can reduce costs by up to 1,408 USD relative to the baseline area. Although a direct linear effect of age is not significant, non-linear patterns emerged, indicating that older individuals may face accelerating costs and that extremely high BMI levels yield diminishing returns in cost increases. Together, these findings point to modifiable risk factors—smoking and obesity management—as prime targets for interventions aimed at lowering overall healthcare expenditures.

To complement these regression insights and further aid in strategic decision-making, I employed classification models to segment individuals into high- and low-cost categories. Among Logistic Regression, Linear Discriminant Analysis, and K-Nearest Neighbors (KNN) approaches, the KNN model (with $K=1$) delivered the strongest performance, achieving an accuracy of approximately 97.8%. Crucially, it exhibited both high sensitivity and specificity, making it exceptionally adept at correctly identifying individuals at elevated risk of incurring substantial medical costs. This capability enables insurers and healthcare providers to pinpoint those who stand to benefit most from targeted interventions, such as smoking cessation programs, obesity reduction efforts, or region-specific healthcare initiatives. By identifying at-risk policyholders in advance,

insurers can tailor coverage options, adjust premiums more equitably, and ultimately foster better long-term health outcomes for their customer base.

While my model perform well, certain limitations remain. Residual diagnostics in the regression analysis suggest that linear assumptions may not fully capture the complexity of medical cost drivers. Variables beyond those included, such as detailed health histories or socioeconomic factors, could refine future models and improve predictive accuracy. Similarly, although the classification models excel within the given dataset, their performance may vary with different populations or healthcare contexts. Exploring more flexible modeling approaches, incorporating additional predictors, or employing more advanced techniques like ensemble learning could enhance both the robustness and applicability of these results.

In conclusion, this project provides a multifaceted view of medical insurance cost prediction by leveraging both regression and classification methods. I identified key drivers, particularly smoking and BMI, that not only inflate individual costs but also help differentiate between high-risk and low-risk populations. These insights are instrumental for insurers looking to fine-tune their pricing models, allocating resources strategically, and offer incentives that encourage healthier lifestyles. Policyholders, in turn, gain clarity on factors that affect their premiums, and healthcare stakeholders can use this information to develop targeted initiatives to lower medical costs. Although there is room for methodological refinement and expanded data collection, the current findings offer a valuable foundation for informed decision-making, risk management, and cost containment in the healthcare insurance industry.

Part V: References

WHO. (2010, May 6). *A healthy lifestyle - WHO recommendations*. Retrieved from World Health Organization: <https://www.who.int/europe/news-room/fact-sheets/item/a-healthy-lifestyle---who-recommendations>